

Birkir Snær Axelsson
Björn Orri Þorleifsson

1. Always check the cables

```
→ ~ ping 130.208.246.98
PING 130.208.246.98 (130.208.246.98) 56(84) bytes of data.
64 bytes from 130.208.246.98: icmp_seq=1 ttl=52 time=5.30 ms
64 bytes from 130.208.246.98: icmp_seq=2 ttl=52 time=5.58 ms
64 bytes from 130.208.246.98: icmp_seq=3 ttl=52 time=4.61 ms
^C
--- 130.208.246.98 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 4.606/5.160/5.578/0.408 ms
→ ~ sudo nmap -sS 130.208.246.98
Starting Nmap 7.92 ( https://nmap.org ) at 2026-08-25 17:58 GMT
Nmap scan report for 130.208.246.98
Host is up (0.0047s latency).
Not shown: 884 filtered tcp ports (no-response), 114 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
5050/tcp   open  mmcc

Nmap done: 1 IP address (1 host up) scanned in 3.55 seconds
→ ~ □
```

(b)

-

5.

```
sudo tcpdump -vAX -i wlp0s20f3 'port 5001'
```

```
→ ~ sudo tcpdump -vAX -i wlp0s20f3 'port 5001'
[sudo] password for birkir:
dropped privs to tcpdump
tcpdump: listening on wlp0s20f3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
19:03:13.622473 IP (tos 0x0, ttl 64, id 38336, offset 0, flags [DF], proto TCP (6), length 60)
    mini-mekka.41208 > 130.208.246.98.complex-link: Flags [S], cksum 0x7e4f (incorrect -> 0xa4
c7), seq 3105210140, win 64240, options [mss 1460,sackOK,TS val 998425072 ecr 0,nop,wscale 7],
length 0
    0x0000:  4500 003c 95c0 4000 4006 26db c0a8 4445  E..<..@.&...DE
    0x0010:  82d0 f662 a0f8 1389 b915 bf1c 0000 0000  ...b.....
    0x0020:  a002 faf0 7e4f 0000 0204 05b4 0402 080a  ....~0.....
    0x0030:  3b82 c1f0 0000 0000 0103 0307                ;.....
19:03:13.627579 IP (tos 0x0, ttl 52, id 0, offset 0, flags [DF], proto TCP (6), length 60)
    130.208.246.98.complex-link > mini-mekka.41208: Flags [S.], cksum 0x627c (correct), seq 14
73383690, ack 3105210141, win 65160, options [mss 1460,sackOK,TS val 3165658389 ecr 998425072,n
op,wscale 7], length 0
    0x0000:  4500 003c 0000 4000 3406 c89b 82d0 f662  E..<..@.4.....b
    0x0010:  c0a8 4445 1389 a0f8 57d2 0d0a b915 bf1d  ..DE....W.....
    0x0020:  a012 fe88 627c 0000 0204 05b4 0402 080a  ....b|.....
    0x0030:  bcb0 1d15 3b82 c1f0 0103 0307                ....;.....
19:03:13.627608 IP (tos 0x0, ttl 64, id 38337, offset 0, flags [DF], proto TCP (6), length 52)
    mini-mekka.41208 > 130.208.246.98.complex-link: Flags [.], cksum 0x7e47 (incorrect -> 0x8d
d5), ack 1, win 502, options [nop,nop,TS val 998425078 ecr 3165658389], length 0
    0x0000:  4500 0034 95c1 4000 4006 26e2 c0a8 4445  E..4..@.&...DE
    0x0010:  82d0 f662 a0f8 1389 b915 bf1d 57d2 0d0b  ...b.....W...
    0x0020:  8010 01f6 7e47 0000 0101 080a 3b82 c1f6  ....~G.....;...
    0x0030:  bcb0 1d15                ....
19:04:35.553365 IP (tos 0x0, ttl 64, id 38338, offset 0, flags [DF], proto TCP (6), length 95)
    mini-mekka.41208 > 130.208.246.98.complex-link: Flags [P.], cksum 0x7e72 (incorrect -> 0xc
0ee), seq 1:44, ack 1, win 502, options [nop,nop,TS val 998507003 ecr 3165658389], length 43
    0x0000:  4500 005f 95c2 4000 4006 26b6 c0a8 4445  E..._@.&...DE
    0x0010:  82d0 f662 a0f8 1389 b915 bf1d 57d2 0d0b  ...b.....W...
    0x0020:  8018 01f6 7e72 0000 0101 080a 3b84 01fb  ....~r.....;...
    0x0030:  bcb0 1d15 5265 6d65 6d62 6572 2074 6f20  ....Remember.to.
    0x0040:  6368 6563 6b20 666f 7220 6e65 7477 6f72  check.for.networ
    0x0050:  6b20 636f 6e6e 6563 7469 7669 7479 0a    k.connectivity.
19:04:35.557912 IP (tos 0x0, ttl 52, id 6331, offset 0, flags [DF], proto TCP (6), length 52)
    130.208.246.98.complex-link > mini-mekka.41208: Flags [.], cksum 0x0d92 (correct), ack 44,
win 509, options [nop,nop,TS val 3165740319 ecr 998507003], length 0
    0x0000:  4500 0034 18bb 4000 3406 afe8 82d0 f662  E..4..@.4.....b
    0x0010:  c0a8 4445 1389 a0f8 57d2 0d0b b915 bf48  ..DE....W.....H
    0x0020:  8010 01fd 0d92 0000 0101 080a bcb1 5d1f  .....].
    0x0030:  3b84 01fb                ;...
```

(c) In your own words, explain what each of the three commands does. Hint: it may help to read the manual (man page) for ncat and tcpdump.

```
sudo tcpdump -vAX -i ens192 'port 5000':
```

Here we have a tcpdump command used to capture and display network traffic on an interface. ran with sudo since capturing raw traffic needs elevated privileges. It is paired with the Options -v for slightly more verbose output. -A which prints each packet's data in ASCII and -X which

additionally prints it in hex. `-i ens192` tells it which interface to listen on and `'port 5000'` filters it down to only traffic on that port.

```
ncat -ln 5000:
```

Here we have a `ncat` command which alone is a command used to read and write data across networks but here it is also paired with the Options `-l` which makes it listen for connections instead of connecting to a remote machine which is the default and `-n` which disables hostname resolution so it skips resolving names to IPs and vice versa rather than requiring one.

```
ncat 130.208.246.98 5000
```

Here the `ncat` command without any Options so it has the Hostname which was disabled in the last command and the port as well which is the default way of using `Ncat`, so this particular command connects to whatever is listening on that IP and port and lets you exchange data with it interactively

d)

```
→ ~ sudo tcpdump -vAX -i wlp0s20f3 'port 5001'
dropped privs to tcpdump
tcpdump: listening on wlp0s20f3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
19:32:50.639327 IP (tos 0x0, ttl 64, id 63197, offset 0, flags [DF], proto UDP (17), length 71)
    mini-mekka.37567 > 130.208.246.98.complex-link: UDP, length 43
    0x0000:  4500 0047 f6dd 4000 4011 c5a7 c0a8 4445  E..G..@.@....DE
    0x0010:  82d0 f662 92bf 1389 0033 7e65 5265 6d65  ...b.....3~eReme
    0x0020:  6d62 6572 2074 6f20 6368 6563 6b20 666f  mber.to.check.fo
    0x0030:  7220 6e65 7477 6f72 6b20 636f 6e6e 6563  r.network.connec
    0x0040:  7469 7669 7479 0a                                tivity.
```

(e) **How does the `tcpdump` output differ when you used UDP vs TCP? Why?**

The main difference is that UDP sends the data in a single packet without establishing a connection, while TCP establishes a connection and uses acknowledgements.

- **UDP:** One packet is sent directly:
 - `proto UDP (17)`
 - Contains the 43-byte message immediately.
 - No handshake or ACK packets.
- **TCP:** Several packets are visible:
 - `SYN` -> starts the connection.

- SYN-ACK -> server accepts it.
- ACK -> completes the TCP three-way handshake.
- PSH-ACK -> sends the n-byte message.
- ACK -> confirms the data was received.

Output

UDP:

19:32:50 IP mini-mekka.37567 > 130.208.246.98: UDP, length 43

"Remember to check for network connectivity"

TCP:

19:33:11 IP mini-mekka.52038 > 130.208.246.98: Flags [S] # SYN

19:33:11 IP 130.208.246.98 > mini-mekka.52038: Flags [S.] # SYN-ACK

19:33:11 IP mini-mekka.52038 > 130.208.246.98: Flags [.] # ACK

19:33:29 IP mini-mekka.52038 > 130.208.246.98: Flags [P.], length 43

"Remember to check for network connectivity"

19:33:29 IP 130.208.246.98 > mini-mekka.52038: Flags [.] , ack 44 # ACK